

Filtering

Filtering means **selecting only the data items that match a condition**.

It is like checking every message in an inbox but opening only messages from one person.

Basic idea

- A dataset contains many items, such as student records, shopping bills, or word cards.
- A condition is a question with a Yes/No answer.
- Check each item one by one.
- If the condition is true, process that item.
- If the condition is false, ignore it and move to the next item.

Example: To find Ahmed's total spending, check every bill. Add the bill amount only if the customer name is Ahmed.

Filtering flow

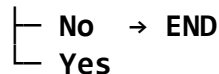
START



Set result to 0



Are there more items?



Pick one item



Does it match the condition?



Check the next item

The result may be a total, a count, or a list.

Common uses

1. Finding a total

To find the total money spent by Ahmed:

- Start with `ahmedSpend = 0`.
- Check each shopping bill.
- If the bill belongs to Ahmed, add its amount to `ahmedSpend`.

2. Count matching items

To count verbs in a paragraph:

- Start with `verbCount = 0`.
- Check each word.
- If the word is a verb, increase `verbCount` by 1.

3. Separate into two groups

To find total marks of boys and girls:

- Start boysTotalMarks = 0 and girlsTotalMarks = 0.
- Check each student record.
- Add the total marks to the correct group based on gender.

Filtering

Filtering means selecting only the data items that **match a condition**.
It is like checking every message in an inbox but opening only messages from one person.

Real Life Analogy: Email Inbox

✉ Priya	Meeting tomorrow
✉ Ahmed	Project report
✉ Ravi	Assignment
✉ Ahmed	Budget sheet
✉ Sneha	Greetings

Open only messages from Ahmed

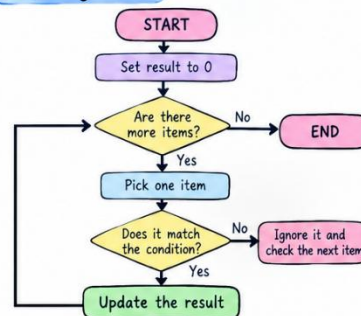
Filtered messages (from Ahmed only)

✉ Ahmed	Project report
✉ Ahmed	Budget sheet

Basic Idea

- 1 A dataset contains many items, such as student records, shopping bills, or word cards.
- 2 A **condition** is a question with a Yes/No answer.
- 3 Check each item one by one.
- 4 If the condition is true, process that item.
- 5 If the condition is false, ignore it and move to the next item.

Filtering Flow



Remember

- The result may be a **total**, a **count**, or a **list**.
- We always check every item in the dataset.
- Only the items that match the condition are used to update the result.

Key Points (Handwritten Notes)

- ✓ Filtering = select items that satisfy a condition.
- ✓ Check each item one by one.
- ✓ Condition gives Yes or No.
- ✓ If Yes → process the item.
- ✓ If No → ignore and move to next.
- ✓ Result can be a **total**, **count** or **list**.

1 Finding a total

To find the total money spent by Ahmed.

Bill No	Customer	Amount (₹)
1	Ahmed	200
2	Priya	150
3	Ahmed	300
4	Ravi	100
5	Ahmed	250

Start: ahmedSpend = 0
Check each bill.
If customer is Ahmed, add the amount.
Final Result: ahmedSpend = 750

2 Count matching items

To count verbs in a paragraph.

Sentence:
She **runs** to school and **plays** with her friends.

Words:

She runs to school and plays with her friends

Start: verbCount = 0
Check each word.
If the word is a verb, increase verbCount by 1.
Final Result: verbCount = 2 (runs, plays)

3 Separate into two groups

To find total marks of boys and girls.

Name	Gender	Total Marks
Rohan	Boy	400
Priya	Girl	450
Amit	Boy	380
Neha	Girl	420

Start: boysTotalMarks = 0
girlsTotalMarks = 0
Check each student record.
Add the total marks to the correct group based on gender.
Final Result: Boys = 780 Girls = 870

Selecting cards: Iterator with filtering

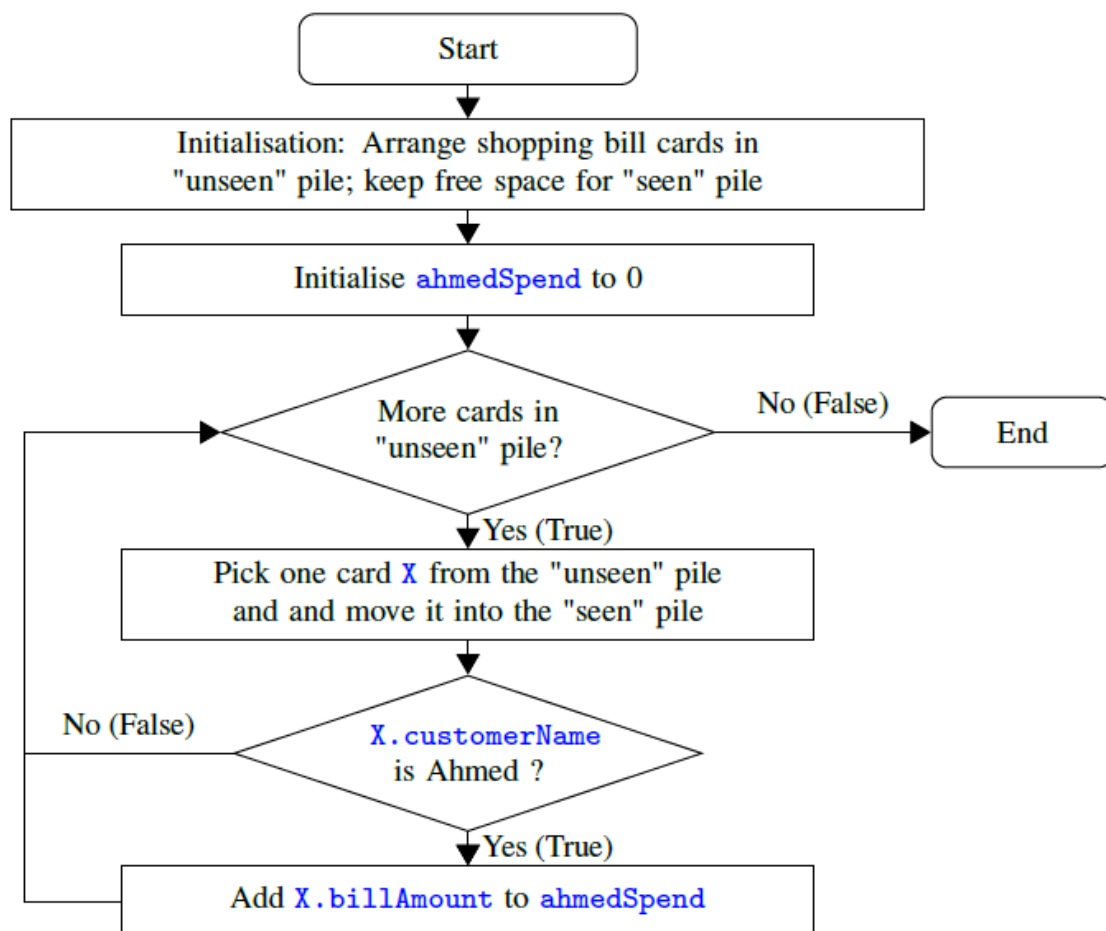
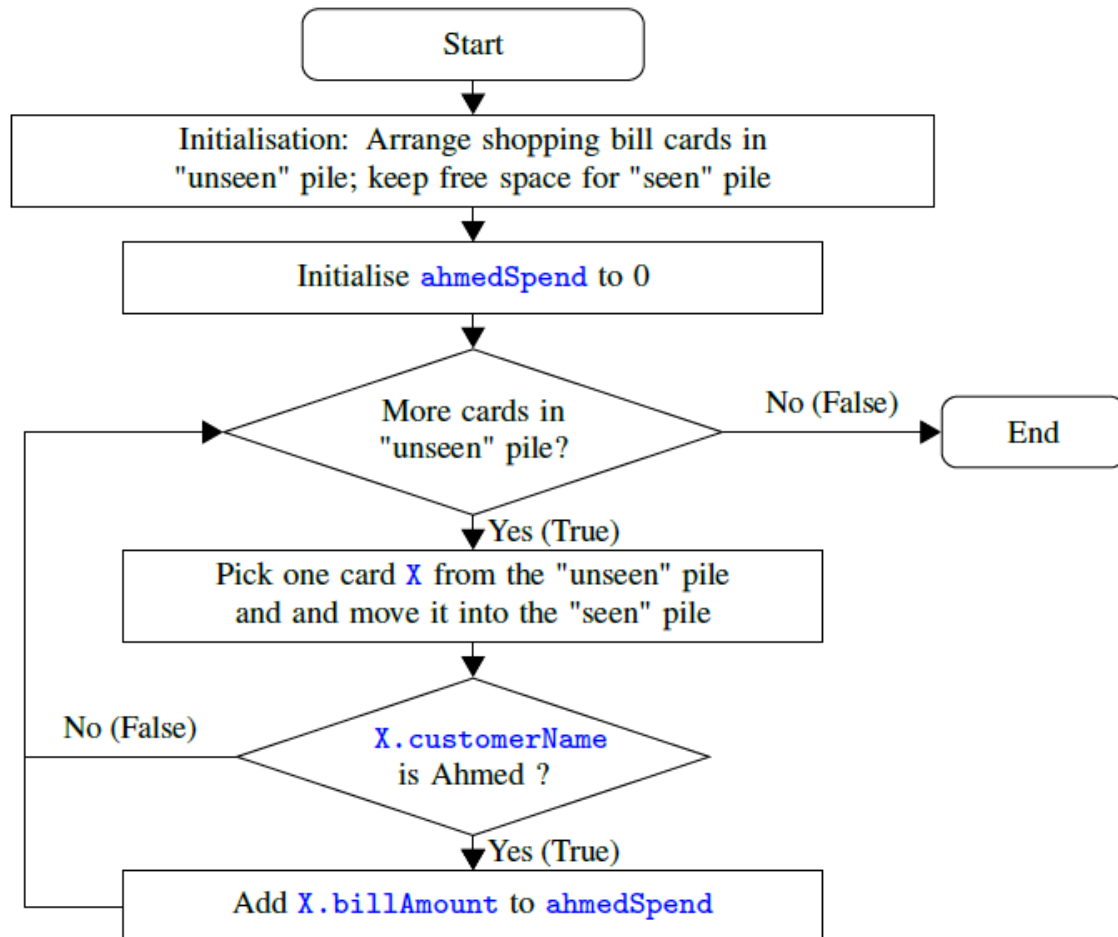


Figure 4.2: Flowchart for finding the total spend of Ahmed

Example:

Suppose we want to find the total spend of one of the customers (lets say Ahmed).

We can do this by examining all the shopping bills, filtering it by the customer name field (checking if the customer name is Ahmed) and then accumulating the bill amounts in an accumulator variable (lets call this variable ahmedSpend). The flowchart for this is shown



Example: Find the count of students belonging to city chennai

	Maths		Physics		Total	
1	Name	Gender	Marks	Marks	Marks	City
2	Anu	F	85	78	163	Chennai
3	Ravi	M	92	88	180	Hyderabad
4	Meera	F	76	84	160	Bengaluru
5	Arjun	M	68	73	141	Chennai
6	Sneha	F	95	91	186	Visakhapatnam
7	Kiran	M	81	79	160	Vijayawada
8	Priya	F	89	94	183	Hyderabad
9	Vikram	M	74	68	142	Bengaluru
10	Divya	F	98	96	194	Chennai
11	Rahul	M	87	82	169	Visakhapatnam
12	Kavya	F	72	77	149	Vijayawada
13	Suresh	M	63	71	134	Chennai
14	Nisha	F	86	89	175	Bengaluru
15	Amit	M	91	93	184	Hyderabad
16	Pooja	F	79	83	162	Visakhapatnam
17	Manoj	M	70	66	136	Vijayawada
18	Lakshmi	F	93	87	180	Chennai
19	Vijay	M	84	90	174	Bengaluru
20	Isha	F	67	75	142	Hyderabad
21	Harish	M	96	89	185	Visakhapatnam

Compound conditions

Sometimes one condition is not enough. A compound condition combines two or more conditions.

AND

AND means every condition must be true.

Example: To add marks of girls from Chennai:

Gender is F AND City is Chennai

Only a record that satisfies both checks is selected.

OR

OR means at least one condition must be true.

Example: To count high scorers:

Maths > 90 OR Physics > 90 OR Chemistry > 90

A student is counted once if they score above 90 in one or more subjects.

Note:

Do not confuse separate checks with one OR condition.

- If you test Maths, Physics, and Chemistry separately, the same student may be counted more than once.
- If you use one OR condition, each matching student is counted only once.

Compound Conditions

Sometimes one condition is not enough. A **compound condition** combines two or more conditions.

It's like applying multiple filters!

AND

AND means **every condition must** be true.

Example: To add marks of girls from Chennai:

Gender is F **AND** City is Chennai

Only a record that satisfies both checks is selected.

Name	Gender	City	Marks	Selected?
Anita	F	Chennai	450	✓ Yes
Priya	F	Delhi	480	✗ No
Rahul	M	Chennai	500	✗ No
Sneha	F	Chennai	470	✓ Yes

OR

OR means **at least one condition** must be true.

Example: To count high scorers:

Maths > 90 **OR** Physics > 90 **OR** Chemistry > 90

A student is counted once if they score above 90 in one or more subjects.

Student	Maths	Physics	Chemistry	Selected?
Arun	95	80	85	✓ Yes
Neha	88	92	70	✓ Yes
Karan	82	78	91	✓ Yes
Divya	89	85	88	✗ No

Note: Do not confuse separate checks with one OR condition.

- If you test Maths, Physics, and Chemistry separately, the same student may be counted more than once.
- If you use one OR condition, each matching student is counted only once.

Finding one item

Sometimes we need only **one** matching item, not all matching items.

Example: Find one student who scored above 90 in Maths, Physics, and Chemistry.

Step-by-step idea

- 1 Set found = False.
- 2 Are there items left AND found = False?
- 3 Check one item.
- 4 Is it the required item?
 - If No → check the next item.
 - If Yes → save it and set found = True.
- 5 Stop when found becomes True or there are no more items.

The variable found is a true/false value used to stop after finding an item.

Real life example:

In a phone contact list, you may want to find one contact with the name "Teacher". As soon as you find it, you can stop checking the rest.

Example: Find one student who scored above 90 in all three subjects

Name	Maths	Physics	Chemistry	Match?
Anil	95	92	94	✓ Found!
Priya	91	85	88	(not checked)
Rahul	96	90	89	(not checked)
Sneha	87	92	95	(not checked)

We stop after finding Anil, so we do not need to check the remaining students.

Stop early when you find the item!

Key terms

- 1 **Filter:** A rule used to select required items.
- 2 **Condition:** A Yes/No check.
- 3 **Variable:** A named place that stores a changing value, such as a count or total.
- 4 **Count:** Number of matching items.
- 5 **Total:** Sum of values from matching items.
- 6 **AND:** All conditions must be true.
- 7 **OR:** At least one condition must be true.
- 8 **Found:** A true/false value used to stop after finding an item.

Finding one item

Sometimes we need only one matching item, not all matching items.

Example: Find one student who scored above 90 in Maths, Physics, and Chemistry.

- Check student records one by one.
- When a matching student is found, save the answer.
- Stop checking further records.

Use a true/false variable named found.

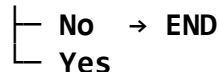
START



Set found = False



Are items left AND found = False?



Check one item



Is it the required item?



└ Yes → Save it and set found = True

Key terms

- **Filter:** A rule used to select required items
- **Condition:** A Yes/No check
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- **Count:** Number of matching items
- **Total:** Sum of values from matching items
- **AND:** All conditions must be true
- **OR:** At least one condition must be true
- **Found:** A true/false value used to stop after finding an item